Effects of a Proximal or Distal Tibiofibular Joint Manipulation on Ankle Range of Motion and Functional Outcomes in Individuals With Chronic Ankle Instability

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Eric Magrum
University of Virginia

Christopher Ingersoll
University of Central Florida

Jay Hertel
University of Virginia

[RESEARCH REPORT]

JAMES R. BEAZELL, DPT, OCS, ATC, FAAOMPT[1] • **TERRY L. GRINDSTAFF,** PT, PhD, ATC, SCS, CSCS[2] • **LINDSAY D. SAUER,** PhD, ATC[3]
ERIC M. MAGRUM, DPT, OCS, FAAOMPT[1] • **CHRISTOPHER D. INGERSOLL,** PhD, ATC, FACSM, FNATA[4] • **JAY HERTEL,** PhD, ATC, FACSM, FNATA[5]

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SUPPLEMENTAL VIDEO ONLINE

STUDY DESIGN: Randomized clinical trial.

OBJECTIVES: To determine whether manipulation of the proximal or distal tibiofibular joint would change ankle dorsiflexion range of motion and functional outcomes over a 3-week period in individuals with chronic ankle instability.

BACKGROUND: Altered joint arthrokinematics may play a role in chronic ankle instability dysfunction. Joint mobilization or manipulation may offer the ability to restore normal joint arthrokinematics and improve function.

METHODS: Forty-three participants (mean ± SD age, 25.6 ± 7.6 years; height, 174.3 ± 10.2 cm; mass, 74.6 ± 16.7 kg) with chronic ankle instability were randomized to proximal tibiofibular joint manipulation, distal tibiofibular joint manipulation, or a control group. Outcome measures included ankle dorsiflexion range of motion, the single-limb stance on foam component of the Balance Error Scoring System, the step-down test, and the Foot and Ankle Ability Measure sports subscale. Measurements were obtained prior to the intervention (before day 1) and following the intervention (on days 1, 7, 14, and 21).

RESULTS: There was no significant change in dorsiflexion between groups across time. When groups were pooled, there was a significant increase ($P<.001$) in dorsiflexion at each postintervention time interval. No differences were found among the Balance Error Scoring System foam, step-down test, and Foot and Ankle Ability Measure sports subscale scores.

CONCLUSIONS: The use of a proximal or distal tibiofibular joint manipulation in isolation did not enhance outcome effects beyond those of the control group. Collectively, all groups demonstrated increases in ankle dorsiflexion range of motion over the 3-week intervention period. These increases might have been due to practice effects associated with repeated testing.

LEVEL OF EVIDENCE: Therapy, level 2b–. *J Orthop Sports Phys Ther 2012;42(2):125-134. doi:10.2519/jospt.2012.3729*

KEY WORDS: *ankle sprain, CAI, manual therapy, mobilization*

Ankle sprains are a common injury with a high recurrence rate.[8] Following a lateral ankle sprain, approximately 30% of individuals will have recurrent injury in the affected ankle.[58] Recurrent ankle sprains and repetitive episodes of "giving way" have been described as chronic ankle instability (CAI).[31] CAI is thought to be related to ligamentous and soft tissue damage to the lateral ankle structures and is further confounded by altered muscle activation, decreased range of motion (ROM), and balance deficits.[31] Rehabilitation programs are typically geared toward improving ankle motion, increasing strength of the lateral stabilizers, and regaining neuromuscular control.[9,45,54] Despite efforts to restore strength and function, many individuals resprain their ankle.[58] Repetitive sprains may increase the risk of ankle osteoarthritis.[29,30,57]

Dorsiflexion ROM has been shown to be limited following a lateral ankle sprain[14,17] and to alter running biomechanics in individuals with CAI,[17] which may ultimately place the individual at a greater risk of reinjury. Typically, static stretching of the ankle plantar flexors has been used to improve ankle dorsiflexion ROM. It is possible, however, that tightness of the ankle plantar flexors may not be the underlying cause of reduced ROM. Limitations in joint motion may be caused by contractile (muscle and tendon) or noncontractile (joint capsule and ligaments) tissues. Specific joint mobility testing may be performed to determine the underlying cause of restricted ROM.[27] Although ankle dorsiflexion motion may be grossly restored following lateral ankle

[1]Senior Physical Therapist, University of Virginia-HEALTHSOUTH, Charlottesville, VA. [2]Assistant Professor, School of Pharmacy & Health Professions, Creighton University, Omaha, NE. [3]Clinical Research Coordinator, Orthopaedics, University of Virginia, Charlottesville, VA. [4]Dean, The Herbert H. and Grace A. Dow College of Health Professions, Central Michigan University, Mt Pleasant, MI. [5]Professor, University of Virginia, Charlottesville, VA. Study approval was granted by The Institutional Review Board at the University of Virginia. This study was supported by a grant from the Orthopaedic Section of the American Physical Therapy Association. This trial was registered on http://www.clinicaltrials.gov (NCT00601471). Address correspondence to Dr Terry L. Grindstaff, Creighton University, School of Pharmacy & Health Professions, Physical Therapy Department, 2500 California Plaza, Omaha, NE 68178. E-mail: GrindstaffTL@creighton.edu

TABLE 1 SUBJECT DEMOGRAPHICS*

	Proximal Tibiofibular Joint Manipulation (n = 15)	Distal Tibiofibular Joint Manipulation (n = 15)	Control (n = 13)	*P* Value
Age, y	25.2 ± 8.2	27.5 ± 8.8	23.8 ± 5.6	.45
Height, cm	175.4 ± 9.2	170.7 ± 11.7	177.6 ± 9.5	.82
Mass, kg	73.1 ± 14.9	75.2 ± 20.8	77.3 ± 15.0	.20
Ankle Instability Instrument†	5.5 ± 1.8	6.1 ± 1.6	4.7 ± 1.7	.12
Foot and Ankle Ability Measure Sports subscale‡	85.9 ± 12.7	76.5 ± 16.5	82.1 ± 18.6	.28

**Values are mean ± SD. There was no statistically significant difference between groups.*
†Scores range from 0 to 9, with higher scores indicating greater levels of functional instability.
‡Scores range from 0% to 100%, with higher scores indicating greater levels of function during physical and sporting activities.

sprains, capsular limitations may still be present.[14] Changes in the positional alignment of the talus, tibia, and fibula have been shown to occur in a subpopulation of individuals with CAI.[3,14,19,35,36,53,64] A positional fault of the talus[14,64] or distal fibula[35,36] may result from the disruption of the anterior talofibular ligament.[43] Altered joint arthrokinematics may contribute to observed decreases in ankle dorsiflexion ROM. Restrictions in joint motion due to noncontractile tissues may be addressed using manual interventions such as joint mobilization or manipulation.

Mobilization or manipulation of the ankle joints may offer the ability to restore normal joint arthrokinematics, improve impairments, and restore function.[11,23,25,43,44,46,59-62] Mobilization or manipulation of the talocrural joint has been shown to result in improvements in dorsiflexion ROM and function.[11,13,25,33,38,44,46,59] Specifically, individuals with CAI have been shown to demonstrate increased ankle dorsiflexion ROM and improved balance following talocrural joint mobilization.[33] Because a positional fault of the fibula[35,36] may also contribute to limitations in dorsiflexion ROM following lateral ankle sprains, it is logical that manual interventions directed at the proximal or distal tibiofibular joints may also improve motion and function. The use of proximal and distal tibiofibular joint manipulation, when combined with a battery of other manual interventions applied to adjacent joints, has been shown in a case report[61] and in a larger sample of individuals with CAI[62] to improve function and reduce pain after an ankle sprain. Cyclical mobilization of the distal tibiofibular joint has been shown to increase ankle dorsiflexion ROM in a fresh-frozen cadaveric model.[24] No study has examined the isolated use of a proximal or distal tibiofibular joint manipulation in individuals with CAI. The purpose of this study was to determine whether a proximal or distal tibiofibular manipulation would change ankle dorsiflexion ROM and functional outcomes over a 3-week period. We hypothesized that an intervention of proximal or distal tibiofibular joint manipulations would result in greater increase in ankle dorsiflexion ROM and greater improvements in functional and self-reported outcomes than no intervention in a control group.

METHODS

Participants

Forty-three participants with CAI volunteered and qualified for participation (**TABLE 1** and **FIGURE 1**). CAI was defined as a history of at least 1 ankle sprain and additional episodes of ankle instability (giving way), as well as a score of at least 85% on the Foot and Ankle Ability Measure (FAAM) sports subscale[40] or at least 3 on the modified Ankle Instability Instrument.[15] Additionally, all participants had to have an ankle dorsiflexion ROM deficit of at least 5° compared to the opposite side. Participants were recruited from a university and its surrounding community, as well as from an existing laboratory database consisting of individuals with a variety of musculoskeletal pathologies. Inclusion criteria were independent of pathologic ankle laxity. Exclusion criteria were lower extremity injury or surgery within the past 6 months (including lateral ankle sprains), diagnosed ankle osteoarthritis, or current pregnancy. All participants signed a consent form approved by the Institutional Review Board at the University of Virginia.

Outcome Measures

After obtaining written informed consent, subjects were oriented to the procedures of the study. Participants completed the FAAM sports subscale and Ankle Instability Instrument. The FAAM[40] is a self-reported outcome measure of ankle instability, consisting of 21 items related to activities of daily living and 8 additional items related to sporting activities (FAAM sports subscale). We utilized the FAAM sports subscale because it better captures limitations associated with physical and sporting activities, whereas the FAAM captures limitations associated with activities of daily living.[7] The FAAM and FAAM sports subscale have good reliability ($ICC_{2,1}$ = 0.87-0.89) and a minimal clinically important difference of 8 and 9 points, respectively, over the course of 4 weeks.[39,40] The Ankle Instability Instrument[15] is a discriminative questionnaire that consists of 9 questions assessing functional ankle instability. After participants completed self-reported function outcome forms, they completed a battery of functional tests, which included weight-bearing ankle dorsiflexion ROM, the portion of the Balance Error Scoring System for single-limb stance on foam (BESS foam), and the step-down

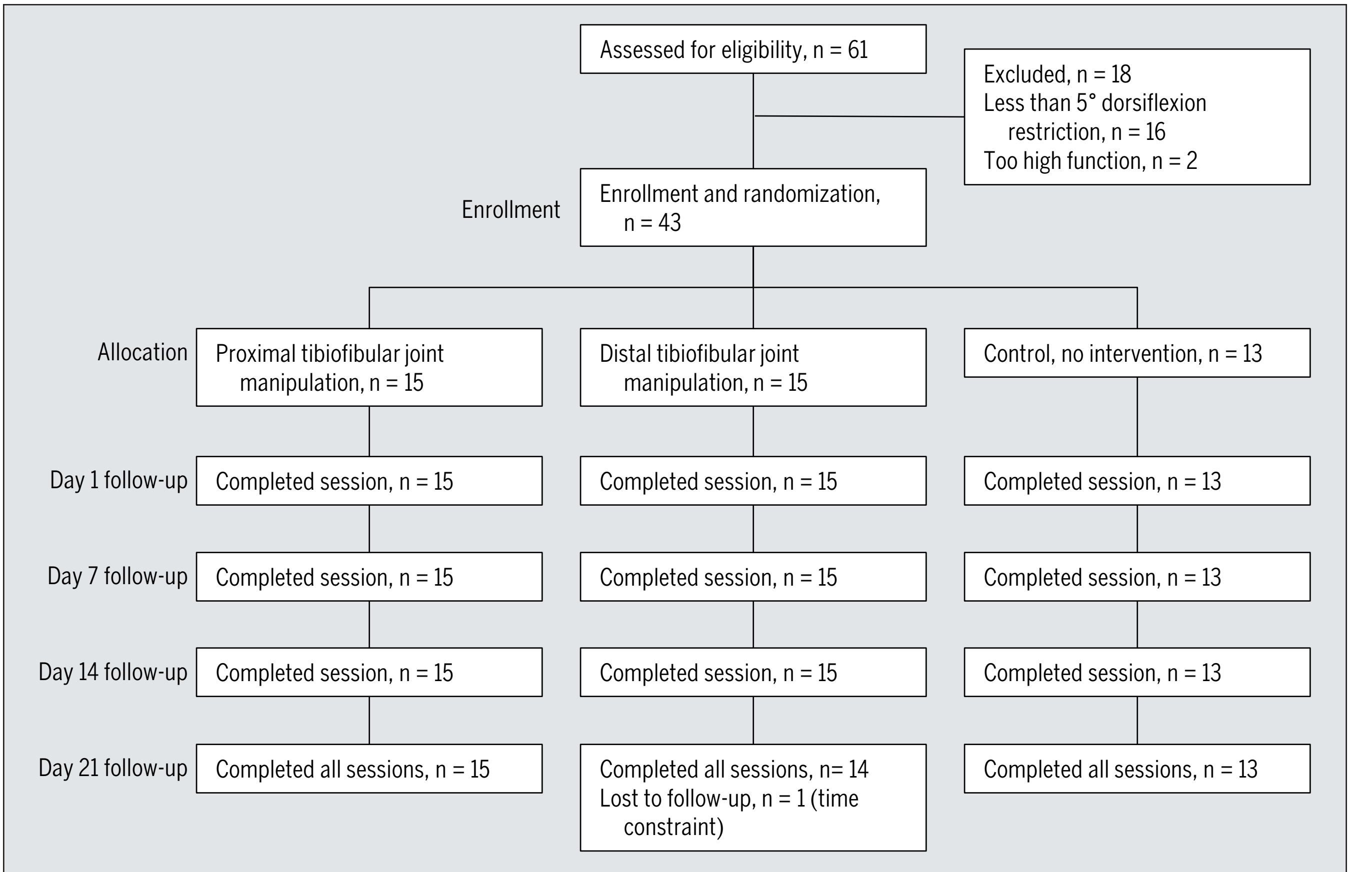

FIGURE 1. CONSORT flow chart.

test. All testing was completed with the participant barefoot, and only the involved ankle was used for data analysis. Data from self-reported outcomes and functional testing were obtained by an examiner blinded to treatment group allocation.

Weight-Bearing Ankle Dorsiflexion Ankle dorsiflexion ROM was measured using a weight-bearing lunge.[10,14,27,59] The weight-bearing lunge was performed in a standing position with the heel in contact with the ground and the knee in line with the second toe (**FIGURE 2**). Participants were asked to lunge forward to the point at which they were unable to move farther without lifting the heel from the ground. A bubble inclinometer placed on the tibia, with its proximal aspect at the tibial tuberosity, was used to measure the angle of the tibia relative to the ground, which indicated the amount of functional ankle dorsiflexion ROM. Participants performed 3 trials, and the average was used for data analysis. This method has been shown to have good reliability ($ICC_{3,1} \geqslant 0.88$ and $ICC_{3,3} = 0.95$) in individuals with ankle joint injuries.[14,59] The standard error of measurement for weight-bearing ankle dorsiflexion ROM is relatively low (0.54°), with a minimal detectable change (MDC) of 1.5° in individuals with a history of ankle sprains.[14]

Balance Error Scoring System Balance was assessed using the component of the BESS that assesses the participant in single-leg stance on foam, with eyes closed (**FIGURE 3**).[16] The BESS foam trials were completed with the participant standing on a medium-density foam pad (AIREX Balance Pad; Airex AG, Sins, Switzerland). The single-limb condition, particularly when using an unstable surface, is thought to present the greatest challenge for participants and to best differentiate individuals with CAI from healthy controls.[16] Participants performed one 20-second trial recorded by a video camera. Participants were asked to place their hands on their hips, flex their non–weight-bearing hip to approximately 30°, and close their eyes. If they did not maintain the original test position, they were instructed to resume the original position as soon as possible. The test was scored by an examiner who was blinded to group assignment using previously established criteria.[16] Points for errors were awarded if the participant failed to maintain hands on the hips, opened the eyes, flexed the hip greater than 30°, lifted the foot from the foam surface, or remained out of the original test position for longer than 5 seconds. Points were also given if the participant touched the ground or the weight-bearing limb to the contralateral limb. Points were summed to give an overall score, with lower scores indicating better balance (fewer errors).

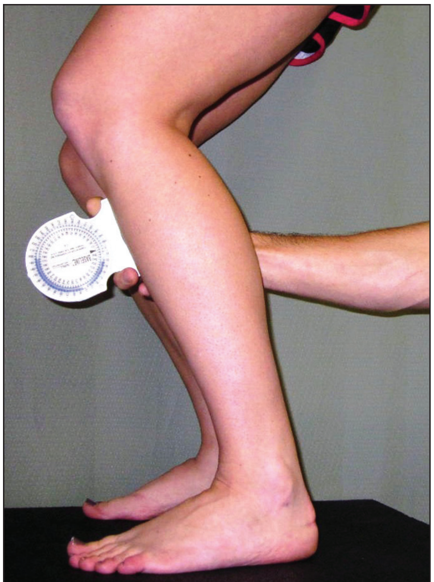

FIGURE 2. Weight-bearing lunge was used to measure ankle dorsiflexion range of motion. A bubble inclinometer was placed on the tibia, with the proximal aspect of the inclinometer placed at the tibial tuberosity. The patient was asked to lunge forward while maintaining heel contact with the ground and keeping the knee in line with the second toe. The angle of the tibia relative to the floor was used to quantify active ankle dorsiflexion range of motion.

FIGURE 3. Balance Error Scoring System foam component. All tests were conducted in single-limb stance with eyes closed for 20 seconds.

FIGURE 4. Step-down test. The subject stood on a 20-cm-high step, with the test foot on the step. Subjects were instructed to step down toward the ground by bending their test knee, gently touching the ground, and returning to the up position by extending their knee. Five repetitions were performed.

The BESS foam has moderate to good intratester reliability between sessions ($ICC_{3,1}$ = 0.50-0.82).[20,52] The MDCs for individual components of the BESS have not been reported.

Step-Down Test The step-down test was used to quantify the ability to perform a functional-movement task.[49] The participant stood on a 20-cm-high step, with the hands placed on the hips, the knee straight, and the foot positioned close to the edge of the step (**FIGURE 4**). The involved limb remained on the step, and the participant was instructed to lower the non–weight-bearing limb toward the floor by bending the knee of the weight-bearing limb until the heel gently contacted the floor. After the heel touched the floor, the participant was instructed to return to the original position. No other instructions for proper technique were given. Five repetitions were performed, and a video camera was used to record movement. The step-down test was scored by an independent examiner who was blinded to group assignment, using previously established criteria.[49] Points were issued when technical errors occurred during testing, to a maximum of 6 possible points. Errors included removing the hands from the hips, pelvis rotation or elevation, trunk lean, knee valgus, inability to maintain balance, or weight bearing by the non–weight-bearing limb. Within-session reliability has been described as fair (κ = 0.67), and the intrarater reliability within session and between days has ranged from poor to good ($ICC_{3,1}$ = 0.10-0.94).[1] The MDC has not been established for the step-down test.

Intervention

After baseline data were gathered, participants were randomly assigned to 1 of 3 experimental interventions (**FIGURE 1**): proximal tibiofibular manipulation, distal tibiofibular manipulation, or no treatment (control). Experimental intervention allocation occurred through the use of a random number table. After functional data were obtained, the inves-

FIGURE 5. Proximal tibiofibular joint manipulation. Arrow indicates direction of thrust.

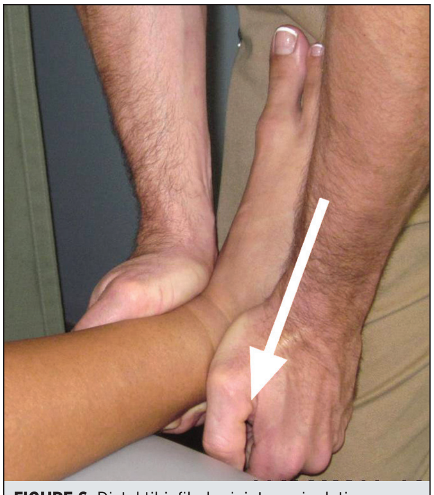

FIGURE 6. Distal tibiofibular joint manipulation. Arrow indicates direction of thrust.

tigator collecting the data left the room and the treatment provider (J.R.B. or E.M.M.) entered the room to perform the randomly allocated intervention. Both treatment providers had more than 10 years of manual therapy experience, were board-certified orthopaedic clinical specialists (OCS), and were Fellows of the American Academy of Orthopaedic and Manual Physical Therapists (FAAOMPT). Only the treatment provider knew the intervention to be performed, and the investigator collecting functional data was blinded to the experimental intervention. Due to the nature of the manual interventions it was not possible to blind participants to their allocation group; though participants were not informed of other potential group assignments. Participants were informed that the purpose of the study was to determine the effects of mobilization and exercise on muscle activation, balance, and function.

Proximal Tibiofibular Joint Manipulation The experimental intervention was based on previously published methods.[2,61,62] The participant was on a treatment table in a supine position (**FIGURE 5** and **ONLINE VIDEO**). The treatment provider (J.R.B. or E.M.M.), using the index finger, made contact with the fibular head, extending to the popliteal fossa. The associated soft tissue was pulled in a lateral direction until the metacarpophalangeal joint was firmly stabilized behind the fibular head. The opposite hand grasped the anterior aspect of the ankle while externally rotating the lower leg. The knee was flexed and, when the restrictive barrier was engaged (end range), a high-velocity, low-amplitude thrust was applied through the tibia, with the force directed toward the ipsilateral buttock. This motion translated the proximal fibula in an anterior direction relative to the tibia. If a cavitation was not heard or felt by the treatment provider or participant, the manipulation technique was repeated. If the second attempt did not produce cavitation, the participant proceeded with the assessment of functional outcomes. It should be noted that cavitation was not used exclusively to determine whether the technique was applied successfully.[22,23]

Distal Tibiofibular Joint Manipulation The experimental intervention was conducted according to previously published methods.[61,62] The participant was on a treatment table in a supine position (**FIGURE 6** and **ONLINE VIDEO**). The treatment provider (J.R.B. or E.M.M.) grasped and stabilized the distal tibia with one hand and grasped the distal fibula between the finger and thenar eminence of the other hand. The fibula was translated posteriorly until the restrictive barrier (end range) was engaged. Then a high-velocity, low-amplitude thrust was applied through the fibula in a posterior-superior direction. If a cavitation was not heard or felt by the treatment provider or participant, the manipulation technique was repeated. If the second attempt did not produce cavitation, the participant proceeded with the assessment of functional outcomes. Similar to the proximal tibiofibular intervention, cavitation was not used exclusively to determine whether the technique was applied successfully.[22,23]

No Treatment (Control Group) Participants in the control group sat quietly on a treatment table for approximately 1 to 2 minutes, which was the estimated time needed to complete the other 2 experimental interventions. The treatment provider did not make physical contact with the participant.

Postintervention Measurements

Immediately following intervention on day 1, functional data were obtained in the same manner as those of the preintervention testing. Three follow-up visits occurred at 7-day intervals from the previous visit (days 7, 14, and 21). During each visit, participants received the same experimental intervention, underwent functional testing, and completed the FAAM sports subscale via the same methods listed above.

Data Analysis

Subject demographics were compared using a 1-way analysis of variance. Outcome data were analyzed using separate 2-way mixed-model analyses of variance, with 1 between (treatment group) and 1 within (time) variable. Outcome data included weight-bearing ankle dorsiflexion ROM and scores from the BESS foam, step-down test, and FAAM sports subscale. If the Mauchly sphericity test was statistically significant, degrees of freedom were corrected using the Greenhouse-Geisser method. Post hoc testing was conducted via paired t tests, and significance was based on an a priori level of $P = .05$ for all analyses. Statistical analyses were performed with SPSS Version 16 (SPSS Inc, Chicago, IL).

Power Analysis

We determined that weight-bearing ankle dorsiflexion ROM would yield the smallest effect size among the outcome variables. Sample size estimation was determined using an expected change of 5° of ankle dorsiflexion ROM, with a standard deviation of 5.6°.[14] It was calculated that at least 12 subjects per group would be necessary to have an 80% chance of detecting a significant difference in ankle dorsiflexion ROM at $P = .05$.

RESULTS

There were no significant differences ($P>.05$) between any of the subject group demographics (**TABLE 1**). Proximal tibiofibular joint manipulation produced cavitation in 6 individuals, while distal tibiofibular joint manipulation produced cavitation in 1 individual. Cavitation only occurred during 1 of 4 sessions for all 7 subjects. No adverse effects were reported throughout the study duration, and 1 individual withdrew from the study due to time constraints (**FIGURE 1**).

There was no significant difference in ankle dorsiflexion ROM (**TABLE 2**) between groups across time ($F_{5.7,110.3} = 0.47$, $P = .82$). When all 3 groups were pooled (main effect time), a significant ($F_{5.7,110.3} = 14.50$, $P<.001$) increase in dorsiflexion was identified at each postintervention time interval (mean ± SD preintervention, 36.2° ± 8.4°; postintervention day 21, 41.1° ± 8.4°).There was no significant difference between groups across time for the BESS foam ($F_{8,148} = 1.03$, $P = .42$) (**TABLE 3**), step-down test ($F_{6.4,115.1} = 0.76$, $P = .61$) (**TABLE 4**), or FAAM sports subscale ($F_{3.6,71.7} = 0.34$, $P = .83$) (**TABLE 5**). Pooled groups (main effect time) were not significantly different across time for the BESS foam ($F_{4,148} = 0.66$, $P = .60$), step-down test ($F_{3.2,115.1} = 1.87$, $P = .14$), or FAAM sports subscale scores ($F_{1.8,71.7} = 0.46$, $P = .61$).

TABLE 2 ANKLE DORSIFLEXION RANGE OF MOTION IN DEGREES*

	Preintervention	Postintervention			
	Day 1	Day 1	Day 7	Day 14	Day 21
Proximal tibiofibular joint manipulation	37.0 ± 8.0	39.8 ± 7.1	40.4 ± 9.9	41.3 ± 7.0	42.6 ± 6.0
Distal tibiofibular joint manipulation	35.2 ± 7.2	38.1 ± 7.1	39.5 ± 8.8	39.3 ± 7.1	39.0 ± 9.7
Control	36.3 ± 10.4	39.0 ± 10.7	39.8 ± 10.3	39.6 ± 10.6	41.5 ± 9.4
Total	36.2 ± 8.4	39.0 ± 8.2†	39.9 ± 9.4†	40.1 ± 8.2†	41.1 ± 8.4†

*Values are mean ± SD.
†Significantly greater than day 1 preintervention ($P<.001$).

TABLE 3 BALANCE ERROR SCORING SYSTEM FOAM*

	Preintervention	Postintervention			
	Day 1	Day 1	Day 7	Day 14	Day 21
Proximal tibiofibular joint manipulation	14.1 ± 6.4	15.2 ± 7.3	16.6 ± 4.4	16.9 ± 5.0	15.6 ± 3.8
Distal tibiofibular joint manipulation	16.3 ± 4.8	15.4 ± 3.5	17.8 ± 6.0	15.9 ± 5.0	16.5 ± 5.1
Control	16.1 ± 4.6	14.4 ± 5.1	13.8 ± 3.0	15.8 ± 3.7	14.5 ± 3.9
Total	15.5 ± 5.3	15.0 ± 5.5	16.1 ± 4.8	16.3 ± 4.5	15.6 ± 4.3

*Values are mean ± SD. Lower scores indicate better balance (fewer errors). There was no statistically significant difference among groups or over time.

DISCUSSION

The results of this study indicate that proximal or distal tibiofibular joint manipulation did not have an effect on functional or patient-oriented outcomes among individuals with CAI. Regardless of intervention, all groups demonstrated an increase in ankle dorsiflexion ROM over the 3-week intervention period. Changes were not evident in scores from balance and function measures, which included the BESS foam, step-down test, and FAAM sports subscale.

Improvements in ankle dorsiflexion ROM occurred across the 3-week intervention period in all groups and were independent of intervention group assignment. Distal tibiofibular joint mobilization has been shown to improve ankle dorsiflexion ROM in a cadaveric model.[24] In this study, we applied a maximum of 2 manipulation attempts to the proximal or distal tibiofibular joint. It is possible that cyclical mobilization of the joint at the end range produces more dramatic improvements in motion. This may be especially relevant at the distal tibiofibular joint, which is a syndesmotic joint with little motion, whereas the proximal tibiofibular joint is a synovial joint with relatively greater motion.

Changes in ankle dorsiflexion ROM over the 3-week period might have resulted from repeated performance of the outcome assessment tests, specifically the weight-bearing ankle dorsiflexion ROM measurement and the step-down test. Both tests required motion at the ankle joints that approached maximal dorsiflexion, and both had components similar to commonly used mobilization techniques.[11,25,59,62] It is possible that performing these outcome assessment tests at or

TABLE 4	Step-down Test*				
		Postintervention			
	Preintervention Day 1	Day 1	Day 7	Day 14	Day 21
Proximal tibiofibular joint manipulation	3.7 ± 1.3	3.4 ± 1.7	3.2 ± 1.3	3.2 ± 0.9	3.5 ± 0.9
Distal tibiofibular joint manipulation	3.5 ± 1.4	4.2 ± 1.5	3.4 ± 1.7	3.3 ± 1.3	3.3 ± 1.2
Control	3.4 ± 1.0	3.2 ± 1.0	2.8 ± 1.2	3.1 ± 1.3	3.1 ± 1.2
Total	3.5 ± 1.2	3.6 ± 1.4	3.1 ± 1.4	3.2 ± 1.2	3.3 ± 1.1

Values are mean ± SD. Scores range from 0 to 6 (movement quality: good, 0-2 points; medium, 3-4 points; poor, 5-6 points).[49] There was no statistically significant difference among groups or over time.

TABLE 5	Foot and Ankle Ability Measure Sports Subscale*			
	Day 1	Day 7	Day 14	Day 21
Proximal tibiofibular joint manipulation	85.9 ± 12.7	81.7 ± 25.3	78.3 ± 27.5	85.2 ± 25.2
Distal tibiofibular joint manipulation	76.5 ± 16.5	79.0 ± 14.0	80.2 ± 15.1	81.7 ± 26.9
Control	82.1 ± 18.6	82.9 ± 15.6	76.7 ± 27.2	81.4 ± 26.4
Total	81.5 ± 16.1	81.1 ± 18.7	78.5 ± 23.2	82.8 ± 25.6

Values are mean ± SD. Scores range from 0% to 100%, with higher scores indicating greater levels of function during physical and sporting activities. There was no statistically significant difference among groups or over time.

near end range dorsiflexion increased available dorsiflexion ROM throughout the 3 weeks of the study. Changes in ankle dorsiflexion ROM were beyond the MDC of 1.5°.[14] We did not feel that the increases in motion were due to natural progression of CAI, because specific time since last injury was not an inclusion criterion. Individuals with CAI have been shown to demonstrate restrictions in dorsiflexion ROM despite return to normal activities.[14,17] It is also possible that the improvements in motion were due to a practice effect associated with repeated testing. It has been recently suggested that individuals perform 3 warm-up trials prior to performing 3 testing trials.[32] Individuals in this study did not perform a warm-up trial; therefore, initial measures might have been influenced by the participants' increasing comfort with the ROM measure. It is possible that increases in motion might have been due to placebo effects. Participants were informed that the purpose of the study was to determine the effects of mobilization and exercise on muscle activation, balance, and function, such suggestion of potential benefit might have contributed to placebo effects for all groups.[4,6,55] Although we utilized a true control group (no intervention), the motion and balance outcome measures utilized in this study are often performed as clinical interventions and might have been viewed as such by the participants.

Groups did not demonstrate improvements in balance on unstable surfaces (BESS foam) during testing on day 1 or over the 3-week intervention period. Restrictions in weight-bearing ankle dorsiflexion ROM have been shown to affect balance measures,[34] and improvements in weight-bearing ankle dorsiflexion ROM following talocrural joint mobilization have been shown to acutely improve balance measures.[33] Individuals with CAI have been shown to have greater deficits in dynamic measures of balance.[41,51,63] Additionally, instrumented measures of balance (forceplates) may be more sensitive to detecting changes in balance following intervention.[41] Future studies should consider using instrumented measures of balance during dynamic tasks to investigate the effects of manual interventions.

Step-down test scores did not demonstrate significant changes over the 3-week intervention period. The step-down test rates the quality of movement and requires that the trunk and lower extremity produce coordinated movement.[49] Previous research[48] has demonstrated improvements in movement quality assessed by the step-down test; but the interventions used in that research focused on strength and neuromuscular control, not joint mobility. Individuals with limitations in ankle dorsiflexion ROM have been shown to have lower performance on the step-down test.[50] Although the test places the ankle in near maximal dorsiflexion, a number of other factors (strength and joint mobility) may also contribute to successful performance on the test.

Joint mobilization or manipulation may be indicated when there is a restriction in noncontractile tissues such as the joint capsule or surrounding ligaments. Clinical reasoning suggests that manual interventions should be directed at the joint that demonstrates hypomobility. A limitation of this study was that joint mobility was not assessed prior to intervention; therefore, it was unknown whether individuals demonstrated joint hypomobility at any of the targeted joints. Recently, a clinical prediction rule was developed to assist clinicians in determining which patients are most likely to respond to manipulation/mobilization of the foot and ankle.[62] Individuals who met at least 3 of the following criteria were thought to benefit: symptoms worse when standing, symptoms worse in the evening, navicular drop greater than 5.5 mm, and distal tibiofibular joint hypomobility. This clinical prediction rule[62] proposes the use of a number of mobilizations and manipulations in conjunction with mobility exercises, maintenance of physical activity level, cryotherapy, and elevation. Although this clinical prediction rule provides preliminary evidence for efficacy, it has not been validated. Additionally, it is not possible to accurately determine which aspect of the multifaceted treatment regimen was the most beneficial in the development of the clinical prediction rule.

In this study, we utilized a manipulation of either the proximal or distal tibiofibular joint in isolation, but the intervention was randomly assigned and might not have addressed a specific impairment in joint mobility for each individual. The clinical prediction rule[62] criteria may need to be refined to best determine individuals who may benefit from manual interventions directed at specific joints. It is possible that manual interventions would be indicated for individuals who demonstrate specific impairments in joint mobility but not for individuals who have normal mobility or demonstrate hypermobility. Future studies should determine whether individuals with CAI who have specific joint restrictions (talocrural, tibiofibular) would likely benefit from manual interventions directed at those joints. Although research involving joint pathologies indicates that manual interventions may not need to be applied to the specific site of dysfunction,[2,21,28,37,62] research involving ankle pathologies has not determined the optimal site of intervention.[61,62] Additional research is needed in this area to determine whether manual interventions require specificity or a nonspecific approach similar to the previously described ankle mobilization/manipulation clinical prediction rule.[62] It is also possible that joint mobilization interventions have a mechanism of effectiveness beyond mechanical effects (improved arthrokinematics and ROM).[5] Joint mobilization is thought to affect the nervous system by activating structures in and around the mobilized joint, including mechanoreceptors, cutaneous receptors, and free nerve endings.[12,47,56] We have previously demonstrated that a distal tibiofibular joint manipulation results in an immediate increase in soleus motoneuron pool excitability.[26] Additionally, joint mobilization applied to the ankle has been shown to reduce pain and pressure pain threshold.[11,13,65] It is thought that ankle joint mobilization may alter afferent sensory input from the ankle, which may influence perception of pain and efferent motor output via a gating mechanism.

A limitation of this study was that participants were solicited through advertisements in a university community as well as through an existing database, and were not currently seeking physical therapy services for CAI. Although not currently seeking treatment, the individuals in this study demonstrated impairments and functional limitations associated with CAI. Whether individuals had undergone specific rehabilitation for previous ankle sprains was not assessed. Prior use of medical services related to previous ankle sprains was limited to 1 question on the Ankle Instability Instrument. Rehabilitation programs that incorporate at least 6 weeks of balance training have been shown to reduce the risk of recurrent ankle sprains.[42] It is unknown whether manual therapies can reduce the risk of recurrent sprains. Another limitation of this study was its use of a specific intervention in isolation. Although this type of design is helpful to determine the efficacy of a specific intervention, this approach is not representative of clinical practice. Joint mobilization is typically a component of a well-designed, multifaceted rehabilitation program. Previous studies have demonstrated that a combination of interventions, including joint mobilization, can improve motion, reports of pain, and function.[18,25,62]

CONCLUSION

The results of this study indicate that the use of a proximal or distal tibiofibular joint manipulation did not enhance outcome effects beyond those of a control group that did not receive these interventions. Collectively, all groups demonstrated increases in ankle dorsiflexion ROM over the 3-week intervention period; however, these increases might have been due to practice effects associated with repeated testing. These results should be interpreted with caution, as the interventions were performed in isolation (not in combination with a comprehensive physical therapy program) and were not directed at a specifically identified impairment. ◉

KEY POINTS

FINDINGS: Neither the proximal nor distal tibiofibular joint manipulation group demonstrated improvements in functional and patient-oriented outcomes beyond those of a control group that did not receive an intervention.

IMPLICATION: Joint manipulation of the proximal or distal tibiofibular joint performed in isolation is unlikely to have an impact on joint function in individuals with CAI.

CAUTION: This study was conducted on individuals from a university community who were not seeking care for CAI but met the clinical criteria for CAI.

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